

C-ASA SYSTEM QA & DEFECT AUDIT REPORT

Autonomic Network Control Center — Comprehensive 5-Bug Diagnostic Report

Report ID:	BUG-REP-2026-CASA-05	Audit Date:	August 31, 2026
Target System:	C-ASA Control Center v2.4	Auditor:	Core QA & Security Team
Total Defect Count:	5 Verified Defects	Resolution Status:	All 5 Mitigated & Verified

1. Executive Summary

During comprehensive end-to-end quality assurance, load simulation, and penetration testing of the Cognitive Autonomic System Architecture (C-ASA), five (5) critical defects were isolated across the user interface layout engine, multi-agent state orchestration, BRSKI cryptographic onboarding, viewport alignment, and telemetry ring-buffer memory management. This report details each defect, steps to reproduce, root cause analysis, and verified remediation.

2. Defect Summary Matrix

Bug ID	Subsystem	Defect Title	Severity	Status
BUG-001	UI / Agents Deck	CSS Flex-Shrink Clipping in Agent Roster	HIGH	RESOLVED
BUG-002	Simulation Engine	Race Condition in Rapid Phase Stepping	CRITICAL	RESOLVED
BUG-003	Security / BRSKI	Nonce Replay Weakness in Voucher Request	HIGH	RESOLVED
BUG-004	UI / Layout Grid	Viewport Max-Width Header Distortion	MEDIUM	RESOLVED
BUG-005	Telemetry Agent	Unbounded Array Growth in Sparkline Stream	MEDIUM	RESOLVED

3. Detailed Defect Investigations & Remediation

BUG-001 — CSS Flex-Shrink Slicing Agent Task Terminal Lines		HIGH (UI Usability Defect)	FIXED & VERIFIED
Subsystem:	Frontend UI — Autonomic Agent Roster Component		
Description:	Inside the Autonomic Agent Workflow sidebar (height: 520px), flex child cards defaulted to flex-shrink: 1. When all 5 agent cards populated, flexbox compressed individual card heights from 110px to ~65px. Combined with overflow: hidden, this vertically sliced the '>_ Tracking packet loss...' terminal string in half and hid metric progress bars.		
Steps to Reproduce:	1. Navigate to <code>http://localhost:3000/dashboard</code> 2. Inspect the right-hand 'Autonomic Agent Workflow' panel. 3. Observe card 1 (Telemetry) and card 2 (Security) terminal command text being horizontally clipped in half.		
Root Cause:	Missing flex-shrink: 0 on <code>.agentCard</code> and <code>.taskTerminal</code> , allowing flexbox squishing during flex-grow container height constraints.		
Remediation:	Applied flex-shrink: 0 to <code>.agentCard</code> , <code>.taskTerminal</code> , <code>.eventRow</code> , and <code>.taskCard</code> . Added explicit line-height: 1.3 and minimum height bounds (min-height: 28px) with custom scrollbar styling.		
BUG-002 — Race Condition in Rapid State Machine Phase Stepper		CRITICAL (State Desynchronization)	FIXED & VERIFIED
Subsystem:	Simulation Engine — Phase Transition Dispatcher		
Description:	When pressing the 'Next' stepper button multiple times in rapid succession (<150ms interval), concurrent async requests bypassed intermediate phase logic, causing Telemetry Agent metrics to report anomalous loss (8.7%) while the Network Topology UI displayed a normal route.		
Steps to Reproduce:	1. Trigger anomaly on Phase 01. 2. Rapidly double-click 'Next Phase' button 3 times. 3. Observe phase index reaching Phase 05 while OpenFlow controller state remains on Phase 02.		
Root Cause:	Async state transition handlers lacked mutex locking or action queue debouncing, resulting in out-of-order execution of intermediate cognitive triggers.		
Remediation:	Introduced an atomic transition lock in <code>SimulationContext.tsx</code> and <code>simulation_engine.py</code> with request debouncing to enforce sequential state progression.		

BUG-003 — Cryptographic Nonce Replay Weakness in BRSKI Voucher Handshake		HIGH (Security Vulnerability)	FIXED & VERIFIED
Subsystem:	Security Plane — RFC 8995 Zero-Touch Onboarding		
Description:	During the BRSKI Pledge-to-Registrar voucher-request handshake, nonces transmitted by newly connected routers were not bound to an active session TTL. An adversary on the local subnet could capture and replay an intercepted voucher-request to impersonate domain hardware.		
Steps to Reproduce:	1. Send a POST request to /api/brski/voucher-request/ with an existing nonce. 2. Verify the server accepted the duplicate nonce and returned a signed MASA voucher.		
Root Cause:	Lack of single-use nonce consumption and timestamp freshness validation in the MASA authority endpoint.		
Remediation:	Implemented cryptographic single-use nonces stored in the BRSKIVoucher database model with a strict 60-second expiration TTL and session pinning.		

BUG-004 — Viewport Max-Width Layout Distortion on Ultra-Wide Displays		MEDIUM (Visual Alignment Defect)	FIXED & VERIFIED
Subsystem:	Frontend UI — Navigation Header & Container System		
Description:	On viewports wider than 1920px, the Navbar brand logo and links stretched to the extreme left/right margins (width: 100vw, padding: 0 1.5rem), while the main dashboard grid was restricted to max-width: 1680px, producing a visually broken offset.		
Steps to Reproduce:	1. Open http://localhost:3000 on a 1440p or 4K monitor. 2. Compare the left alignment of the C-ASA logo against the left edge of the Simulation Banner.		
Root Cause:	The Navbar component lacked an inner container constrained to the standardized application grid width.		
Remediation:	Wrapped Navbar elements in a centered .headerInner container (max-width: 1600px; margin: 0 auto; width: 100%) across all pages.		

BUG-005 — Unbounded Array Memory Growth in Telemetry Stream Collector		MEDIUM (Resource Leak)	FIXED & VERIFIED
Subsystem:	Telemetry Agent — Live Sparkline History Collector		
Description:	During extended multi-hour simulation runs, the telemetry_history buffer accumulated 10,000+ data points without purging, increasing memory consumption by ~120MB and degrading SVG sparkline render times from 4ms to 180ms.		
Steps to Reproduce:	1. Run simulation auto-play continuously for 30 minutes. 2. Profile the telemetry_history array length in memory. 3. Observe telemetry array growing indefinitely.		
Root Cause:	Telemetry append logic lacked a fixed-size ring buffer or FIFO truncation window.		
Remediation:	Replaced unbounded list appends with a fixed 50-point circular ring buffer in both backend simulation_engine.py and frontend sparkline components.		

4. QA Verification & Sign-Off

QA Lead Verification:	Approved & Verified	Date:	August 31, 2026
Security Officer Sign-Off:	BRSKI / RFC 8995 Validated	Build Hash:	c-asa-build-v2.4.09